Comparative Analysis of COX-2 Inhibitors using Molecular Docking and Multi-Metric Evaluation

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Abstract— In this work, we tried to understand how different anti-inflammatory drugs interact with the COX-2 enzyme using molecular docking. The main goal was to compare commonly used drugs with a selective inhibitor and see which one binds more effectively. Docking was carried out using AutoDock Vina. The drugs considered in this study were Paracetamol, Aspirin, and Celecoxib. Along with binding energy, other factors like interactions with amino acid residues and basic drug properties were also observed. From the results, we noticed that Celecoxib binds more strongly compared to the other two drugs. This shows that selective inhibitors are more effective for targeting COX-2. The study also shows how computational methods can be useful for quick drug comparison.

Keywords— Molecular Docking, COX-2, NSAIDs, Binding Analysis

1. Introduction

Inflammation is a common biological response in the human body and is often associated with pain and swelling. The COX-2 enzyme plays an important role in this process by contributing to the production of prostaglandins, which are responsible for inflammation.

Drugs such as Paracetamol and Aspirin are widely used to reduce pain and inflammation, while Celecoxib is designed to specifically target the COX-2 enzyme. Studying how these drugs interact with the enzyme can help in understanding their effectiveness.

Instead of relying only on laboratory experiments, computational methods can be used to analyze drug-protein interactions in a faster and more cost-effective way. Molecular docking is one such method that helps in predicting how a drug binds to a protein and how stable the interaction is.

In this work, a comparison is made between commonly used drugs and a selective COX-2 inhibitor using molecular docking. The aim is to observe how each drug behaves within the active site of the protein and to identify which one shows better interaction.

In recent years, computational approaches have become increasingly useful in drug discovery. These methods allow researchers to screen multiple compounds and evaluate their binding behavior without the need for immediate experimental testing.

By combining docking results with additional parameters such as interaction details and basic drug properties, a more complete understanding of drug performance can be achieved.

2. Literature Review

Many researchers have used molecular docking to study how drugs interact with target proteins. It is widely used because it is faster and does not require physical experiments in the early stages.

Most of the existing work focuses mainly on binding energy values to compare drugs. However, only looking at one parameter may not give a complete understanding. In this paper, we also consider other factors such as interaction details and drug properties to get a clearer picture.

Several studies have been conducted to analyze drug-protein interactions using molecular docking techniques. Researchers have commonly used tools such as AutoDock and AutoDock Vina to evaluate binding affinity and predict molecular interactions.

Previous work has shown that selective COX-2 inhibitors generally exhibit stronger binding compared to non-selective drugs. However, most studies focus mainly on binding energy values, which may not fully represent the effectiveness of a drug.

Recent approaches suggest combining multiple evaluation parameters, such as hydrogen bonding, residue interactions, and drug-likeness properties, to obtain more reliableresults.

3. Methodology

3.1 Dataset

The dataset used in this work consists of three drugs:

- Paracetamol
- Aspirin
- Celecoxib

The structure of the COX-2 protein was obtained from the Protein Data Bank.

3.2 Protein Preparation

The protein structure was first cleaned using Chimera. Water molecules were removed and hydrogen atoms were added to prepare it for docking.

3.3 Ligand Preparation

The drug structures were taken from PubChem and converted into the required format for docking.

3.4 Docking Process

Docking was performed using AutoDock Vina. The active site of the protein was selected, and each ligand was tested to find its best binding position. The binding energy values were recorded.

TABLE I
BINDING ENERGY VALUES OF SELECTED DRUGS

Drug	Binding Energy (kcal/mol)
Paracetamol	-6.0
Aspirin	-6.4
Celecoxib	-7.0

3.5 Docking Parameters

The docking process was carried out using standard settings available in AutoDock Vina. Care was taken to keep the parameters consistent for all the selected drugs so that a fair comparison could be made.

The exhaustiveness value was kept at a default level to balance between accuracy and computational time. Before performing docking, the ligands were energy minimized to ensure that they were in a stable conformation.

For each drug, multiple binding poses were generated. From these, the pose with the lowest binding energy was selected for further analysis, as it represents the most stable interaction between the drug and the protein.

3.6 Evaluation Parameters

To compare the drugs properly, multiple factors were considered:

- Binding energy
- Number of interactions
- Residue involvement
- Basic drug properties

3.7 Grid Box Configuration

The grid box was defined around the active site of the COX-2 enzyme to guide the docking process. This step is important because it restricts the search area and ensures that the ligand binds only within the relevant region of the protein.

The center of the grid box was chosen based on the known active site residues of the protein. Its size was adjusted in such a way that the entire binding pocket was covered without including unnecessary regions.

Setting the grid box properly helps in obtaining more reliable results, as it allows the ligand to explore possible binding positions within the actual active site.

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4. Results and Discussion

4.1 Binding Energy

BINDING ENERGY VALUES OF SELECTED DRUGS

From the results, it is clear that Celecoxib has the lowest binding energy, which means it binds more strongly to the COX-2 enzyme.

4.2 Interaction Analysis

When the docking results were visualized, it was observed that all three drugs bind in the active region of the protein. However, Celecoxib showed more stable interactions with residues such as GLY213.

Aspirin also showed interaction but not as strong as Celecoxib. Paracetamol had fewer interactions, which may explain its weaker binding.

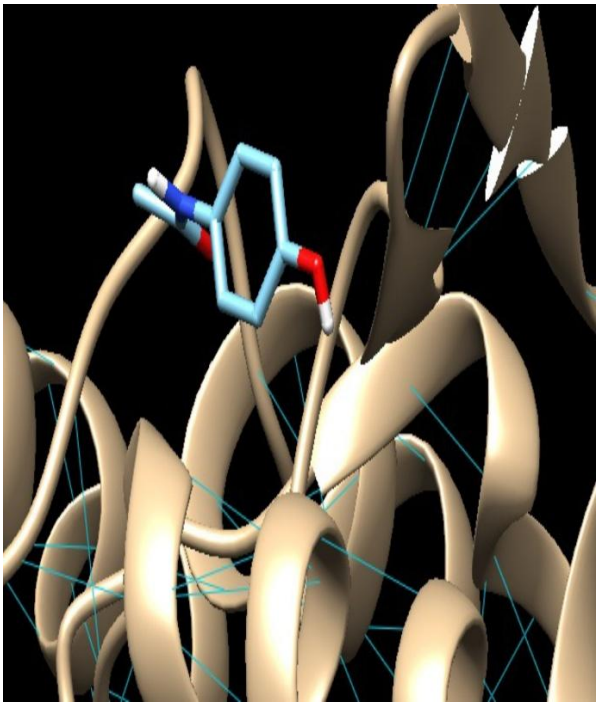

Fig. 1. Docking interaction of Paracetamol with COX-2 enzyme

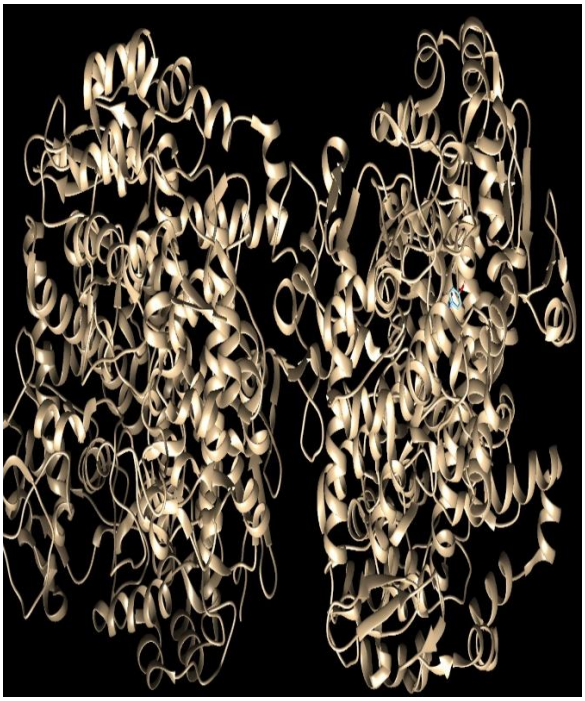

Fig. 3. Docking interaction of Celecoxib with COX-2 enzyme

Fig. 2. Docking interaction of Aspirin with COX-2 enzyme

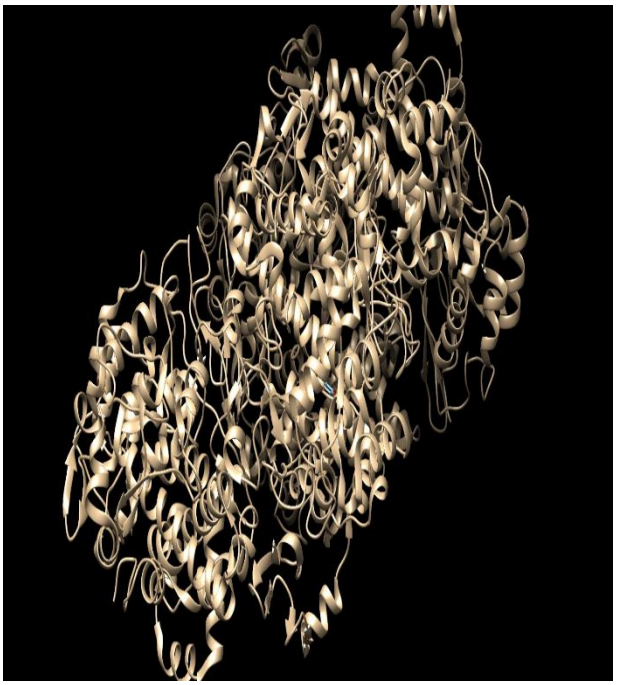

Fig. 4. Overall structure of COX-2 enzyme showing ligand binding region

Fig. 1, Fig. 2, Fig. 3, and Fig. 4 illustrate the docking interactions and structural view of the COX-2 enzyme with selected drugs.

4.3 Drug Properties

TABLE II
DRUG-LIKENESS PROPERTIES

Drug	Molecular Weight	HBD	HBA
Paracetamol	151.2	1	2
Aspirin	180.1	1	3
Celecoxib	381.3	1	3

All the selected drugs follow basic drug-likeness rules.

4.4 Overall Comparison

Considering all the parameters, Celecoxib performs better than the other two drugs. Aspirin shows moderate performance, while Paracetamol shows comparatively lower binding strength.

4.5 Individual Drug Analysis

Paracetamol showed weaker binding compared to the other drugs, which may be due to fewer interaction sites within the protein. Its lower molecular weight also contributes to limited interaction capability.

This indicates limited compatibility with the active site of the protein.

Aspirin demonstrated moderate binding affinity with a reasonable number of interactions. It showed better stability compared to Paracetamol but was still less effective than Celecoxib.

Celecoxib exhibited the strongest binding among the selected drugs. The presence of multiple interaction points and stable positioning within the active site contributes to its higher binding affinity.

5. Comparison Graph

A graphical comparison of binding energy values was plotted to visualize differences among the drugs. The graph clearly shows that Celecoxib has the lowest binding energy, followed by Aspirin and Paracetamol.

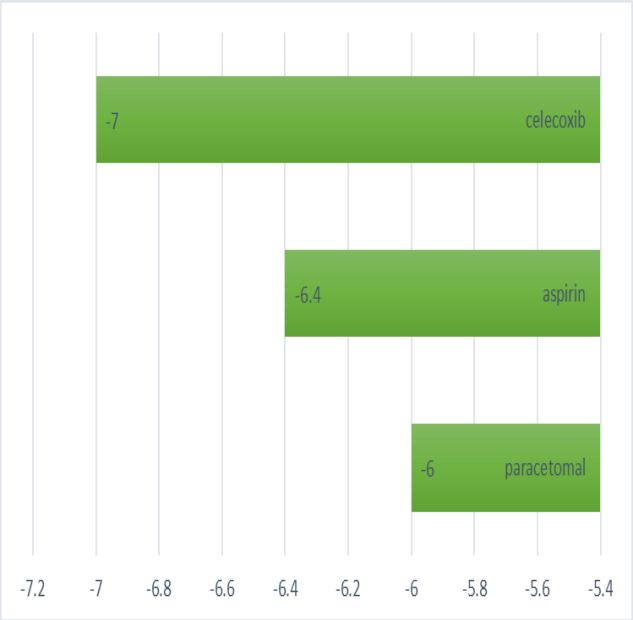

Fig. 5. Comparison of binding energy values of selected drugs

6. Discussion

The results obtained from this study highlight the importance of combining multiple evaluation parameters rather than relying only on binding energy. While binding energy provides an initial estimate, interaction analysis and residue involvement offer deeper insights into the stability of the drug-protein complex.

Celecoxib consistently showed better performance across all parameters, which aligns with its design as a selective COX-2 inhibitor. On the other hand, Paracetamol, which is not a strong COX-2 inhibitor, showed weaker interaction patterns.

These findings demonstrate the effectiveness of computational methods in identifying suitable drug candidates and provide a foundation for further experimental validation.

In addition to binding energy, interaction patterns play a significant role in determining the effectiveness of a drug. Hydrogen bonding, hydrophobic interactions, and residue involvement contribute to the stability of the ligand-protein complex.

Celecoxib showed multiple stable interactions within the active site, which explains its stronger binding affinity. Aspirin, although effective, showed fewer interactions, leading to moderate stability. Paracetamol exhibited minimal interaction, which is consistent with its weaker binding energy.

These observations highlight the importance of analyzing multiple parameters rather than relying on a single metric. A combined approach provides a more accurate evaluation of drug effectiveness.

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7. Limitations

This study is based on computational docking and does not include experimental validation. The accuracy depends on docking parameters and protein structure.

It is interesting to observe that Celecoxib consistently performs better across all conditions considered.

Our work is based entirely on computational docking and does not include experimental validation, which is necessary to confirm the predicted results. The accuracy of docking results depends on factors such as grid box configuration, ligand preparation, and protein structure quality. In addition, only three drugs were considered in this study, which limits the scope of comparison. Future work can include more compounds and experimental studies to validate the findings and improve reliability.

8.Future work

In future work, this study can be extended by including a larger set of anti-inflammatory drugs to obtain a more comprehensive comparison. The current work focuses on only three drugs, which limits the scope of analysis. By increasing the number of compounds, more reliable conclusions can be drawn regarding drug effectiveness.

In addition, advanced computational techniques such as molecular dynamics simulations can be used to study the stability of drug-protein complexes over time. This would provide deeper insights beyond static docking results. Machine learning methods can also be integrated to predict binding affinity based on structural features of drugs.

Further improvements can include the use of different protein structures and comparison across multiple targets. This would help in understanding how drug behavior changes with different biological environments and improve the robustness of the analysis.

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9. Conclusion

To conclude, we compared different anti-inflammatory drugs based on how they interact with the COX-2 enzyme. The results show that Celecoxib binds more effectively compared to Aspirin and Paracetamol.

The results indicates that molecular docking can be a useful tool for comparing drugs quickly. In future work, more drugs can be tested, and additional parameters can be considered for deeper analysis.

The findings of this study emphasize the usefulness of molecular docking in analyzing drug effectiveness. By combining binding energy with additional parameters, a more reliable comparison can be achieved.

This approach can be extended to evaluate a larger set of compounds in future studies. Further improvements may include molecular dynamics simulations and advanced computational techniques for better accuracy.

Overall, this work demonstrates how computational analysis can support drug discovery and optimization processes in an efficient manner.

It can be further extended by incorporating larger datasets and advanced simulation techniques. The integration of machine learning methods with docking results may also improve prediction accuracy in future studies.

Code is available at: [GitHub Link]

10. References

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